PG No. 32 $O_{\rm h}$ $m\bar{3}m$ [cubic]

* character table

$O_{\rm h}$	$1(1)$	$2_{001}(3)$	$2_{110}(6)$	$3^+_{111}(8)$	$4^+_{001}(6)$	$-1(1)$	$\mathrm{m}_{001}(3)$	$\mathrm{m}_{110}(6)$	$-3^+_{111}(8)$	$-4^+_{001}(6)$
A_{1g}	1	1	1	1	1	1	1	1	1	1
A_{2g}	1	1	−1	1	−1	1	1	−1	1	−1
E_g	2	2	0	−1	0	2	2	0	−1	0
T_{1g}	3	−1	−1	0	1	3	−1	−1	0	1
T_{2g}	3	−1	1	0	−1	3	−1	1	0	−1
A_{1u}	1	1	1	1	1	−1	−1	−1	−1	−1
A_{2u}	1	1	−1	1	−1	−1	−1	1	−1	1
E_u	2	2	0	−1	0	−2	−2	0	1	0
T_{1u}	3	−1	−1	0	1	−3	1	1	0	−1
T_{2u}	3	−1	1	0	−1	−3	1	−1	0	1

* polar ↔ axial conversion

A_{1g} (A_{1u}) A_{2g} (A_{2u}) E_g (E_u) T_{1g} (T_{1u}) T_{2g} (T_{2u}) A_{1u} (A_{1g}) A_{2u} (A_{2g}) E_u (E_g) T_{1u} (T_{1g}) T_{2u} (T_{2g})

* symmetric product

	A_{1g}	A_{2g}	E_g	T_{1g}	T_{2g}	A_{1u}	A_{2u}	E_u	T_{1u}	T_{2u}
A_{1g}	A_{1g}	A_{2g}	E_g	T_{1g}	T_{2g}	A_{1u}	A_{2u}	E_u	T_{1u}	T_{2u}
A_{2g}		A_{1g}	E_g	T_{2g}	T_{1g}	A_{2u}	A_{1u}	E_u	T_{2u}	T_{1u}
E_g			$A_{1g}+E_g$	$T_{1g}+T_{2g}$	$T_{1g}+T_{2g}$	E_u	E_u	$A_{1u}+A_{2u}+E_u$	$T_{1u}+T_{2u}$	$T_{1u}+T_{2u}$
T_{1g}				$A_{1g}+E_g+T_{2g}$	$A_{2g}+E_g+T_{1g}+T_{2g}$	T_{1u}	T_{2u}	$T_{1u}+T_{2u}$	$A_{1u}+E_u+T_{1u}+T_{2u}$	$A_{2u}+E_u+T_{1u}+T_{2u}$
T_{2g}					$A_{1g}+E_g+T_{2g}$	T_{2u}	T_{1u}	$T_{1u}+T_{2u}$	$A_{2u}+E_u+T_{1u}+T_{2u}$	$A_{1u}+E_u+T_{1u}+T_{2u}$
A_{1u}						A_{1g}	A_{2g}	E_g	T_{1g}	T_{2g}
A_{2u}							A_{1g}	E_g	T_{2g}	T_{1g}
E_u								$A_{1g}+E_g$	$T_{1g}+T_{2g}$	$T_{1g}+T_{2g}$
T_{1u}									$A_{1g}+E_g+T_{2g}$	$A_{2g}+E_g+T_{1g}+T_{2g}$
T_{2u}										$A_{1g}+E_g+T_{2g}$

* anti-symmetric product

A_{1g}	A_{2g}	E_g	T_{1g}	T_{2g}	A_{1u}	A_{2u}	E_u	T_{1u}	T_{2u}
–	–	A_{2g}	T_{1g}	T_{1g}	–	–	A_{2g}	T_{1g}	T_{1g}